

Data Analysis report for FoodCorp

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Section 1: The KPIs

KPI Description : 1)No of monthly active customers during a specific period of time.

KPI formula: count(unique customers), per month, per store, based on last receipt

Steps to realise KPI:

1) Create via SQL query

```
SELECT store_code,
DATE_TRUNC('month', purchased_at) AS purchase_month,
COUNT(DISTINCT receipts.customer_id) AS unique_customers
FROM receipts
INNER JOIN (
    SELECT customer_id, MAX(receipt_id) AS latest_receipt
    FROM receipts
    GROUP BY customer_id
) AS wow_receipts
ON receipts.customer_id = wow_receipts.customer_id
AND receipts.receipt_id = wow_receipts.latest_receipt
GROUP BY store_code, purchase_month
ORDER BY store_code, purchase_month;
```

- 2) Visualised via Tableau as 'Active Customers'
- 3) Data underpins figure 1 in Comparative Analysis

Additional Notes: None

KPI Description : 2)Active Repeat Customers

KPI formula: count(unique month from purchased_at), per customer, per store

Steps to realise KPI:

- 1) Create via SQL query

```
SELECT customer_id, store_code, COUNT(DISTINCT DATE_TRUNC('month', purchased_at))  
AS active_months
```

```
FROM receipts
```

```
GROUP BY customer_id, store_code
```

```
HAVING COUNT(DISTINCT DATE_TRUNC('month', purchased_at)) > 1;
```

- 2) Visualised via Tableau as Active Repeat Customers
- 3) Data underpins figure 2 in Comparative analysis

Additional Notes: None

KPI Description : 3) First Time Customers

KPI formula: count(unique customer_id), per month, per store

Steps to realise KPI:

- 1) Create via SQL Query

```
SELECT customer_id, store_code, MIN(DATE_TRUNC('month', purchased_at)) as first_timers
```

```
FROM receipts
```

```
GROUP BY customer_id, store_code
```

```
HAVING COUNT(DISTINCT DATE_TRUNC('month', purchased_at)) = 1;
```

- 2) Visualised via Tableau as First Time Customers
- 3) Data underpins figure 3 in Comparative analysis

Additional Notes: None

KPI Description : 4) Busiest Day in a week in each store

KPI formula: count / count(unique day of week from purchased at), per store

Steps to realise KPI:

- 1) Create SQL query

```
SELECT store_code,
```

```
DATE_PART('dow', purchased_at) AS day_of_week,
```

```
COUNT(*) AS purchases_count
```

```
FROM receipts
```

GROUP BY store_code, DATE_PART('dow', purchased_at)

ORDER BY store_code, purchases_count DESC;

- 2) Visualised via Tableau as Busiest Day of the week
- 3) Data underpins figure 4 in Comparative analysis

Additional Notes: None

KPI Description : 5) Time Between Orders

KPI formula: date difference(last_purchase, first_purchase), per customer

Steps to realise KPI:

- 1) Create SQL Query

```
SELECT r1.customer_id,  
  
       MIN(r1.purchased_at) AS first_purchase,  
  
       MAX(r2.purchased_at) AS last_purchase,  
  
       DATEDIFF(MAX(r2.purchased_at), MIN(r1.purchased_at)) AS time_between_orders  
  
FROM receipts r1  
  
JOIN receipts r2 ON r1.customer_id = r2.customer_id  
  
WHERE r2.purchased_at > r1.purchased_at  
  
GROUP BY r1.customer_id  
  
HAVING DATEDIFF(MAX(r2.purchased_at), MIN(r1.purchased_at)) > 0;
```

- 2) Visualised via Tableau as Time between orders

3) Data underpins figure 5 in Comparative analysis

Additional Notes: We have just taken time between first and last purchase so as to help us gauge the future churn rates.

KPI Description : 6) Churn Rate

KPI formula: count(unique customer_id) with no purchase made in the next quarter, per quarter, per store

Steps to realise KPI:

1) Create SQL Query

```
SELECT
YEAR(purchased_at) AS year,
CASE
    WHEN MONTH(purchased_at) BETWEEN 1 AND 3 THEN 'Q1'
    WHEN MONTH(purchased_at) BETWEEN 4 AND 6 THEN 'Q2'
    WHEN MONTH(purchased_at) BETWEEN 7 AND 9 THEN 'Q3'
    ELSE 'Q4'
END AS quarter,
store_code,
COUNT(DISTINCT customer_id) AS churned_customers
FROM receipts
WHERE YEAR(purchased_at) IN (2021, 2022)
GROUP BY YEAR(purchased_at),
```

CASE

WHEN MONTH(purchased_at) BETWEEN 1 AND 3 THEN 'Q1'

WHEN MONTH(purchased_at) BETWEEN 4 AND 6 THEN 'Q2'

WHEN MONTH(purchased_at) BETWEEN 7 AND 9 THEN 'Q3'

ELSE 'Q4'

END,

store_code

ORDER BY year, quarter, store_code;

- 2) Visualised via Tableau as Churn rate
- 3) Data underpins figure 6 in Comparative analysis

Additional Notes: None

KPI Description : 7) Cohort Analysis

KPI formula: count(unique customers), per cohort year, cohort month, store code

Steps to realise KPI:

- 1) Create SQL Query

SELECT

YEAR(first_purchase_date) AS cohort_year,

MONTH(first_purchase_date) AS cohort_month,

r.store_code,

COUNT(DISTINCT CASE WHEN MONTH(r.purchased_at) = MONTH(first_purchase_date)
THEN r.customer_id END) AS cohort_size,

```
    COUNT(DISTINCT CASE WHEN TIMESTAMPDIFF(MONTH, first_purchase_date,  
r.purchased_at) = 1 THEN r.customer_id END) AS active_customers_next_month,
```

```
    COUNT(DISTINCT CASE WHEN TIMESTAMPDIFF(MONTH, first_purchase_date,  
r.purchased_at) = 2 THEN r.customer_id END) AS active_customers_two_months_later,
```

```
    COUNT(DISTINCT CASE WHEN TIMESTAMPDIFF(MONTH, first_purchase_date,  
r.purchased_at) = 3 THEN r.customer_id END) AS active_customers_three_months_later
```

```
FROM (
```

```
    SELECT
```

```
        customer_id,
```

```
        MIN(purchased_at) AS first_purchase_date,
```

```
        store_code
```

```
    FROM receipts
```

```
    GROUP BY customer_id, store_code
```

```
    HAVING YEAR(MIN(purchased_at)) = 2021 AND MONTH(MIN(purchased_at)) = 1
```

```
) AS first_purchases
```

```
JOIN receipts AS r ON r.customer_id = first_purchases.customer_id AND r.store_code =  
first_purchases.store_code
```

```
GROUP BY cohort_year, cohort_month, r.store_code
```

```
ORDER BY cohort_year, cohort_month, r.store_code;
```

- 2) Visualised via Tableau as Cohort Analysis
- 3) Data underpins figure 7 in Comparative analysis

Additional Notes: The above code is executed for 3 months which can be adjusted to view cohorts for further months.

Comparative Analysis

Introduction: In data analysis the Structured Query Language (SQL) and Key Performance Indicators (KPI's) form a powerful duo for effective data analysis. KPI's help measure progress and success in an organisation's data. It gives a good reveal of the current systems and how much progress is being made. KPI's help you to focus on what matters most. It helps you gain hidden insights which can be hidden in large datasets. It helps you track trends in data and helps you analyse them. Business decisions are made easier with the use of KPIs. SQL KPI's bridge the gap between raw data and actionable insights

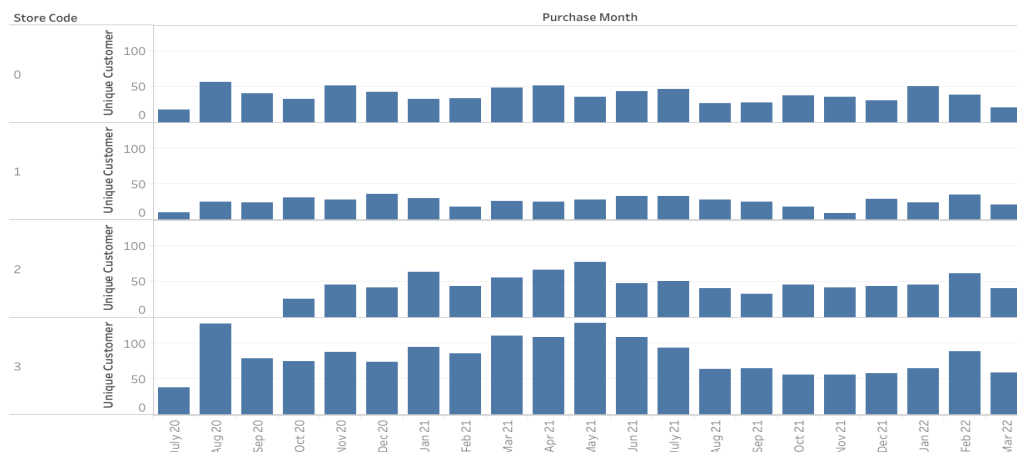
Problem Statement: FoodCorp wants to boost marketing in one of its four stores across the country. Instead of blindly picking the biggest, they're looking for the store with the most potential for explosive growth from the campaign. The organisation is aware that every region has unique shopping habits, preferences, and even perceptions of their brand. So, they're diving deep into customer data to understand who's buying what, where, and why. This intel will guide their marketing and distribution strategies, ensuring the campaign resonates with the right people in the right place. The key factor is to consider regional differences, customer demographics and existing shopping habits in each store to make a wise decision as to which store must be focused on during the next campaign as they only can focus on one store.

KPI Significance :

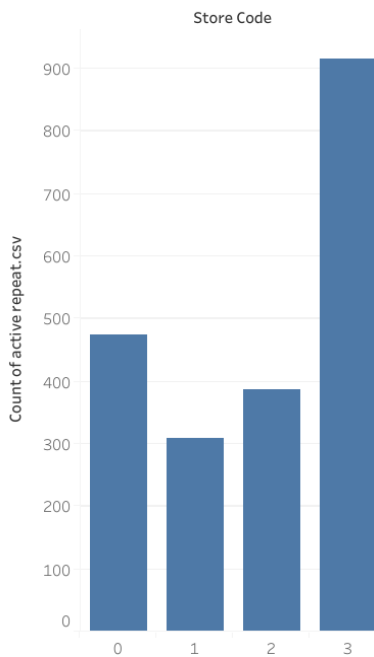
1) Number of monthly active customers during a specific period of time:

This KPI is selected to gauge the customer base of the organisation. It helps us understand whether or not our customer base is growing or declining over time. With this KPI you can understand which store is able to get a higher active customer count and which store is doing relatively poorly. This helps you understand your customer response to marketing efforts and products in store. Store 1 is having the least customers while Store 3 has the maximum customers.

Monthly Active Customers



Active Repeat Customers



2) Active Repeat Customers :

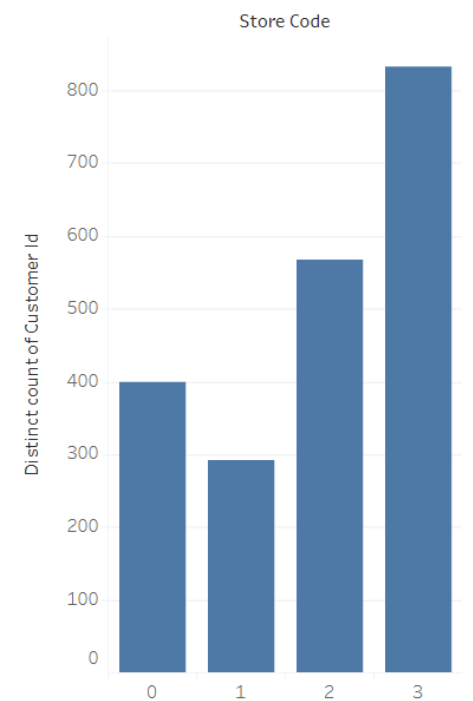
This KPI is used to understand the most valuable customer. It measures the customers that frequently visit time and again expressing their loyalty to the brand. It is also likely that repeat customers will spend more in stores. Higher the rate of repeat customers indicates a strong customer loyalty and satisfaction. Store 1 does the poorest and Store 3 does the best in terms of active customers.

3) First Time Customers :

This KPI helps in understanding the customer acquisition and gauging the market. This mainly focuses on how new customers are attracted to the store and which store was able to draw the attention of customers. This helps in identifying potential markets based on the purchases made by new customers.

Once again we see a similar trend in the stores. Store 1 attracts the least amount of customers and Store 3 the most customers.

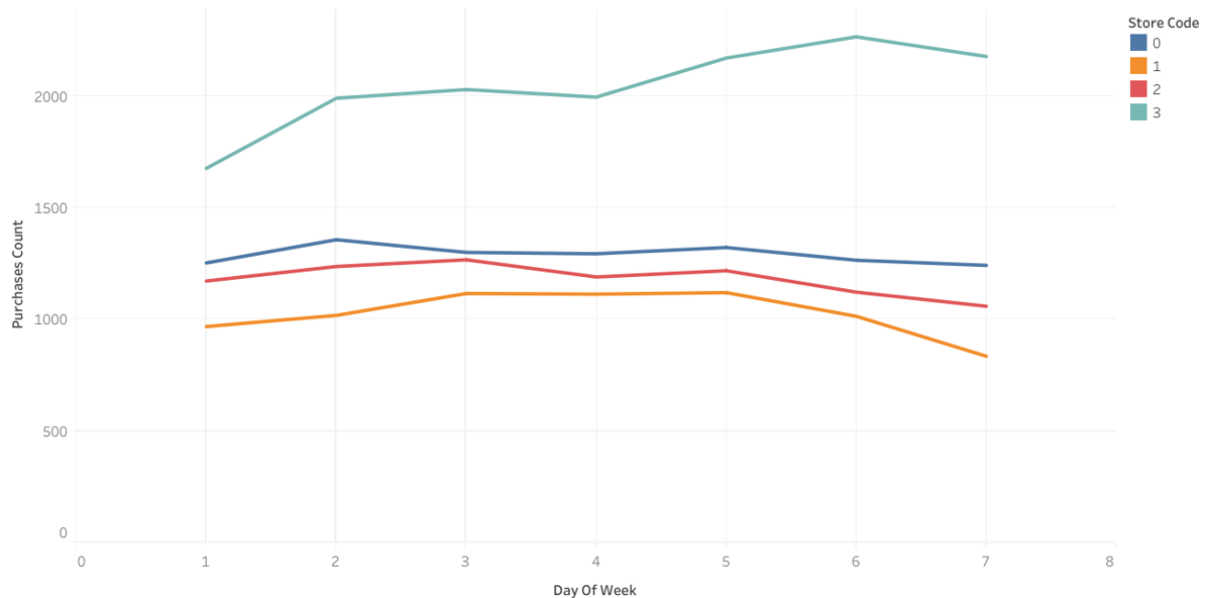
First Time Customers



4) Busiest day in a week in each store :

This KPI is needed to understand the staff requirement in each store so that there is ample staff during busy hours and less staff when things are slow. This improves labour costs and lets management know that for maximum customer satisfaction you must have more staff during busy hours. It also is useful for inventory management so stores can stock up a day before the busy hour hits. The trend seems similar as Store 3 tends to have busier purchase counts during the week and Store 1 tends to be lacking behind.

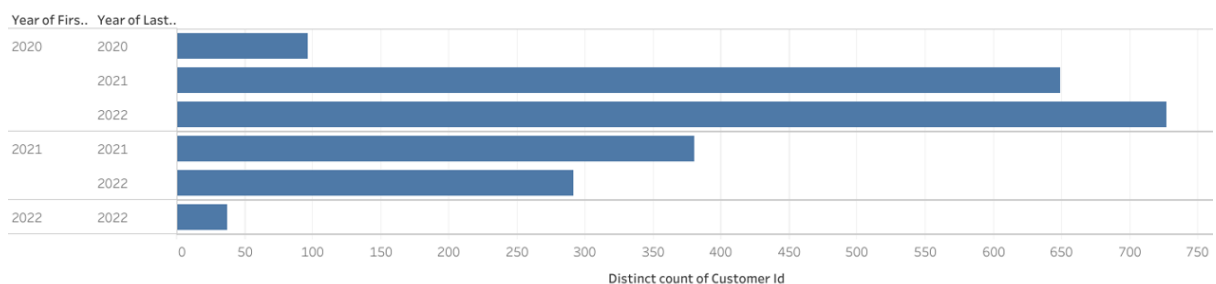
Busiest Day of the Week



5) Time between orders :

This KPI is used mainly to predict future purchases and have a good grip over inventory management. Also helps one identify the potential churn risk.

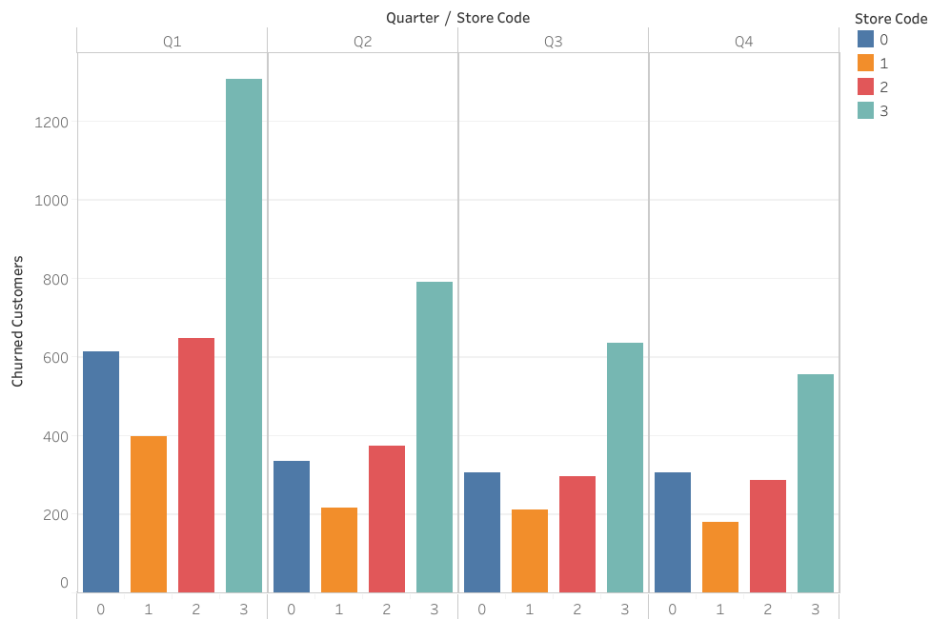
Time between Orders



6) Churn Rate :

It is a KPI that tells you how many customers stop engaging with the organisation over a period of time. It is useful to understand where resources are being used and what kind of profit is seen as an outcome. It is a good way to measure the effectiveness of your efforts and campaigns. Here we see a completely different trend where store 3 churns maximum customers while Store 1 churns the least amount of customers.

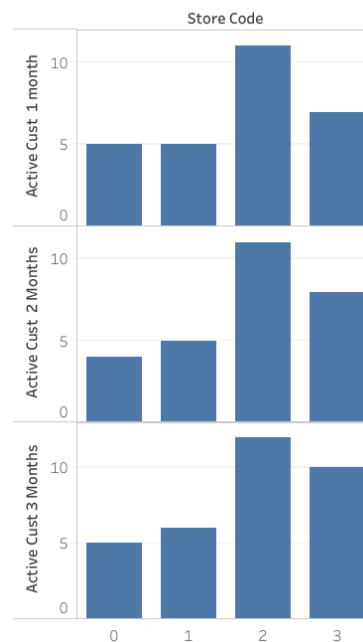
Churn Rate



7) Cohort Analysis:

This KPI is needed to understand customer behaviour. Identification of which customers are retained over a period of months is something that this command achieves. It also helps the organisation understand the potential growth of customers and which branch is doing better than the others. It is heavily used to compare the growth of your organisation with the competitors organisation and understand who was able to retain maximum customers. Here store 2 and 1 has a consistency of customers through the months while store 3 is increasing with time and Store 1 decreasing.

Cohort Analysis



Business Recommendations :

As per the observations made in the above 7 KIPs we notice that Store 1 does very poorly when it comes to attracting new customers and getting a high customer retention. Though Store 2 and Store 0 do comparatively better they still need to adopt methods that Store 3 uses

to boost their active customers and in return boost sales. The busy hours in Store 3 is the maximum and is consistent almost everyday indicating a strong customer base. Store 2 and 0 do a fair job given the active monthly customers. However we do observe that Store 1 does poorly due to the overall customers being very less. When it comes to churn rate we notice that Store 3 churns most customers and Store 1 the least. This might look in favour of Store 1 however since there are not many new customers in Store 1 on a monthly basis it is a very biased view. On the other hand Store 3 churns the maximum customers but also has a high rate of customer sales and customer retention in cohort analysis. Store 1 on the other hand underperforms in cohort analysis with the retention rate dropping as months go by. After careful consideration it is my business recommendation for FoodCorp to increase their marketing spend on **STORE 1**. Since Store 3 is doing fairly well not much work needs to be done to that. Store 0 and 2 are doing good as compared to the poor performance of Store 1. Store 1 although doing bad can use other stores as an example and adopt their success practices. It can also use the new marketing spend to attract new customers and in general boost sales in the future. With the help of the marketing spend we can aim to boost the monthly active customers and the active repeat customers in Store 1.